

Does This Button Work?

Investigating YouTube’s Ineffective User Controls

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Abstract

This paper presents a large-scale experimental audit of YouTube’s user control mechanisms for managing unwanted video recommendations. Drawing on crowdsourced data from 22,722 participants using a custom-built browser extension, we analyzed over 567 million recommendations over six months. The extension introduced a “Stop Recommending” button overlaying recommended videos, which – depending on randomized assignment – triggered one of four native feedback signals to YouTube (e.g., “Dislike,” “Not Interested,” “Don’t Recommend Channel,” “Remove from History”) or no signal at all in the control group. This design allowed us to assess the effectiveness of different user controls through actual user behavior and downstream changes in recommendations. Using a machine learning model trained to estimate video similarity, we quantified how often unwanted content reappeared after user feedback. We find that YouTube’s feedback mechanisms are largely ineffective: even the most “definitive” controls prevented fewer than half of similar recommendations. Since unwanted recommendations are relatively rare, our large-scale approach was essential to detect these effects. These findings reveal a substantial gap between user expectations and platform behavior. We conclude with design and policy recommendations to enhance user agency, transparency, and researcher access for platform accountability.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

recommendation systems; YouTube; crowdsourced data; algorithmic transparency; experimental audit

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1 Introduction

With over two billion active users worldwide, YouTube is currently the second most visited website on the Internet, eclipsed only by its parent company Google’s primary search engine [37]. As part of its core functionality, YouTube employs a sophisticated recommendation system that is responsible for the majority of its video views – surpassing both channel subscriptions and direct search traffic [11]. Over the last decade, a wealth of scholarly and popular media reports has documented concerns about this recommendation algorithm, including claims that it promotes gradual radicalization – often referred to as “rabbit holes” [44] – as well as political extremism [26] and ideological bias [15, 34].

Despite public statements by YouTube executives asserting that the platform optimizes for “user satisfaction” and “time well spent” rather than mere watch time [11, 51], evidence suggests YouTube continues to prioritize engagement metrics over user well-being [28, 40]. A 2019 paper written by Google engineers describes how the algorithm is designed to balance user engagement with user satisfaction [52], and official blogposts describe interventions to demote low-quality or dangerous content and favor “authoritative” sources [11, 13, 49], but the precise policies remain opaque.

To further allow users to customize their experience on YouTube, the platform provides multiple forms of feedback: a Dislike button on every video, a Don’t Recommend Channel option and a Not Interested option in each recommended video, and finally a Remove From Watch History button in every user’s watch history. Although YouTube maintains that users can exert some influence over their recommendations through these feedback tools, anecdotal accounts and previous studies paint a different picture. The Mozilla Foundation’s 2021 YouTube Regrets campaign, for example, highlighted numerous instances in which individuals continued to receive unwanted content – even after proactively taking steps that YouTube suggests for avoiding such material [28].

Poorly designed or nonfunctional user controls can leave individuals feeling disempowered. In collaboration with Mozilla, the non-profit Simply Secure conducted a focused inquiry into YouTube's controls in 2021, concluding that the platform's design does not effectively uphold user-centric principles of freedom, transparency, or control. Their study, entitled "Dark Patterns in User Controls: Exploring YouTube's Recommendation Settings", indicates that YouTube's interface is reactive rather than proactive. Users must play "catch-up" with the recommendation engine, while opportunities to "teach" the algorithm by explicitly shaping its outputs remain sparse and limited [39].

The present study aims to systematically evaluate the efficacy of YouTube's controls by leveraging a large community of Mozilla's RegretsReporter volunteers. 22,722 participants contributed detailed usage data, enabling the creation of a dataset encompassing more than 567 million recommended videos. To our knowledge, this is the largest independent experimental audit of YouTube's recommendation system to date.

1.1 Related research

Research on online recommender systems, particularly those deployed by large-scale social media platforms such as YouTube, has steadily grown over the past decade. This body of work broadly emphasizes three intersecting domains: (1) the technical underpinnings of recommendation algorithms, (2) the sociotechnical implications of personalization and engagement metrics, and (3) user agency and control mechanisms in algorithmic environments.

1.1.1 Technical Foundations of Recommender Systems. Many contemporary online platforms rely on large-scale, deep learning-based recommender systems designed to increase user engagement and retention. YouTube's own recommendation system leverages watch history, user demographics, and contextual session data to generate "next up" videos and personalized feeds [6]. These data-hungry pipelines were originally lauded for their ability to filter through vast libraries of content efficiently, but academics and activists have repeatedly raised concerns about the emphasis on "watch time" or "time spent" mentioned above. Researchers have noted how design choices such as weighting watch time more than user satisfaction may inadvertently promote extreme, radicalizing, or controversial content [30, 36, 44]. Critically, these recommendations are shaped by proprietary algorithms that remain opaque to both end users and external researchers, hindering accountability and transparency.

At the same time, many scholars have also pointed out that the content that users consume on a platform is the output of a complex feedback loop between the user's preferences and choices, the platform's design, its recommender system (which receives the input from the UI), and the content available on the platform at the time [10, 19, 26, 29, 31, 35, 38]. Due to the complex nature of this interaction, it is not trivial to separate the impact of each factor on which content is actually delivered to users.

1.1.2 Sociotechnical implications: engagement, rabbit holes, and algorithmic amplification. A growing literature critiques platforms like YouTube for facilitating "rabbit holes" of increasingly narrow or extremist content [34, 44]. Early investigations documented YouTube's role in radicalizing users by recommending alt-right,

conspiratorial, or otherwise inflammatory videos [21, 26]. Political scientists and media scholars find that YouTube's personalized feeds may create feedback loops reinforcing biased or sensational content, especially if user interactions (clicks, likes, longer watch times) signal the algorithm to serve similar content again [33, 41]. Although YouTube has publicly claimed it optimizes for "user satisfaction" and "time well spent" [11, 50, 51], critics argue that the top priority remains watch time, driving deeper engagement at the potential cost of user well-being and content quality [42].

1.1.3 User agency, control mechanisms and platform design. At the heart of many controversies surrounding YouTube is the question of user agency: To what extent can individuals exercise meaningful control over the algorithmic content they see? A related strand of research highlights "dark patterns," or user interface designs that guide people toward particular outcomes – often those advantageous to the platform rather than the user [12]. Simply Secure's 2021 study provided a design-focused lens on how YouTube's feedback mechanisms (e.g., "Not interested", "Don't recommend channel", "Dislike" etc.) can be confusing, buried, or only marginally effective [39]. Such "dark patterns" can impede user autonomy, hinder transparency, and ultimately diminish trust in the platform [45].

Moreover, scholars investigating the usability of recommender systems argue that platforms often lack intuitive means for user feedback to meaningfully reshape recommendations [20, 23]. Instead, platforms typically rely on inferred preferences (e.g., watch time) rather than explicit signals (e.g., direct feedback). The result is a persistent tension: users want mechanisms to "teach" the algorithm their preferences, yet the available controls may be too limited or poorly implemented to enact real changes in recommendation.

1.1.4 Transparency, Accountability, and Public Policy. Several strands of literature highlight the importance of platform accountability and independent audits [24]. Researchers, journalists, and civil society organizations attempting to study YouTube's algorithm often face steep challenges, including restrictive terms of service and limited access to relevant data [7]. The emergence of crowdsourced studies reflects a creative attempt to overcome these barriers, turning to volunteers to donate usage data for large-scale audits [3, 16, 22, 28]. These studies work their way through data protection regulation (e.g. GDPR), hard-to-find data download options on platforms, and additional layers of anonymity, in order to protect citizen scientists contributing to the project [17]. It is worth pointing out that these studies are not unique to Europe and North America: data donation studies have also been conducted in countries such as India [9] or Pakistan [8], China [48] and Brazil [2].

Additionally, new regulatory frameworks such as the EU's Digital Services Act (DSA) and debates around platform governance underscore the need for improved data access, mandatory transparency, and protections for public-interest research [4, 25]. Many scholars support the idea of "algorithmic accountability" mandates that would require platforms to provide vetted researchers with structured, privacy-preserving access to data on recommendation systems [1]. Such policy interventions aim to shine a light into the "black box" of proprietary algorithms and facilitate more robust public-interest investigations.

In sum, existing literature highlights significant gaps between user expectations of control and the realities of recommendation

algorithms – gaps that appear exacerbated by user interface constraints and opaque system design. Evidence suggests that users’ attempts to manage their recommendations rarely translate to genuine control, especially when tools are reactive, poorly explained, or systematically undervalued by the underlying algorithm. This study thus builds on prior work by offering a large independent audit of YouTube’s user control mechanisms, and calling for stronger algorithmic transparency, better user interface design, and clearer legal protections to enable rigorous, large-scale platform research.

1.2 Research Contribution

Existing literature on YouTube’s recommendation system has largely focused on the system’s role in “rabbit holes,” ideological echo chambers, and content moderation struggles, while tangentially addressing user control mechanisms [34, 44]. However, most prior work relies on secondary data or small-scale user surveys, limiting our understanding of how and to what extent people can actively shape and manage their own recommendation feeds. Moreover, studies that do examine user feedback features often base their conclusions on platform-provided documentation or on anecdotal insights rather than systematic testing [39]. As a result, researchers, policymakers, and designers lack granular, real-world evidence on the efficacy of YouTube’s feedback tools – particularly when users attempt to block or dislike content they find objectionable.

Our study addresses these gaps in two key ways. First, by conducting one of the largest independent audits of YouTube’s recommendation system – drawing on crowdsourced data from 22,722 Mozilla RegretsReporter participants – we provide an unprecedented scale of user-level, real-world evidence. Second, our randomized controlled design tests the effectiveness of each user feedback pathway (e.g., “Dislike,” “Don’t recommend channel,” “Remove from watch history,” etc.) against a control group, allowing us to isolate which controls meaningfully reduce unwanted recommendations. In contrast to YouTube’s native interface, our study uses a custom “Stop Recommending” button that sent one of several feedback signals to YouTube, depending on the participant’s randomly assigned experiment group. Importantly, video rejection events were relatively rare – roughly 30,000 videos were rejected across more than 22,000 users. This highlights the importance of our large-scale design: small datasets would likely miss meaningful patterns in such sparse signals. In doing so, this work contributes empirical clarity to a critical area of research – user agency on large platforms – and directly informs design and policy recommendations for improving transparency, accountability, and user autonomy.

2 Methods

2.1 Recruitment and Consent Process

Participants were recruited through Mozilla’s “YouTube Regrets” project¹. When users installed the RegretsReporter browser extension, they were directed to an onboarding page that explained the purpose of the study, the types of data to be collected, and the experimental nature of the “Stop Recommending” button. This page made it clear that participants might be randomly assigned to a control group in which the button would have no effect on YouTube’s

recommendation system. No research data were collected before explicit consent was obtained. Users who declined participation were not enrolled in the experiment, and their interactions with the button sent a default “Dislike” signal to YouTube without logging any data. The onboarding material is shown in the Appendix. The project was reviewed and discussed extensively by Mozilla’s Trust and Legal teams, and the browser add-on underwent Mozilla’s formal extension review process, which included strengthening the opt-in flow for participants. Finally, after the experiment, we updated the extension to display to all participants a notification with a thank you message, along with a link to a published report.

2.2 Experiment Design

Participants who consented were randomly assigned to one of six experimental arms. Each arm determined what signal (if any) would be sent to YouTube when the participant clicked the custom “Stop Recommending” button (and whether that button was shown):

- **Control:** No feedback sent to YouTube
- **Dislike:** Sends a “Dislike” signal to YouTube
- **Not Interested:** Sends a “Not Interested” signal
- **Don’t Recommend Channel:** Sends a “Don’t Recommend Channel” signal
- **Remove from History:** Removes the video from the user’s watch history
- **UX Control:** Button not shown

The “Stop Recommending” button was visible whenever video recommendations appeared: on the YouTube homepage, the sidebar, and on watched video pages. In all contexts, the button appeared on all recommended videos. Participants in all experimental arms, including control, were free to use YouTube’s native feedback buttons (e.g., “Dislike,” “Not Interested”). This was by design, to measure real-world conditions rather than constrain user behaviour. This possibly attenuates estimated treatment effects.

2.3 Data Collection

The extension collected the following data from consented users:

- A unique anonymous installation ID
- Assigned experiment arm
- Records of all “Stop Recommending” button presses, including timestamps and video IDs
- Records of all YouTube recommendations served (homepage, sidebar, etc.)
- Records of YouTube-native controls used (e.g., Dislike)

No personally identifying information was collected. Data were stored and processed in accordance with Mozilla’s privacy policy.² We excluded a very small number of anomalous participants whose activity was inconsistent with plausible human use (e.g., feedback submissions several orders of magnitude above the 99.9th percentile, likely indicating automated behavior). While this process risks occasional false positives or negatives, including or excluding these outliers does not meaningfully change any reported results.

¹<https://www.mozillafoundation.org/en/youtube/regrets/>

²See Mozilla’s RegretsReporter privacy notice: <https://foundation.mozilla.org/youtube/regretsreporter/privacy-notice/>

Table 1: Videos in each class in the labeled dataset.

Classification	Percentage
Acceptable Recommendation	75% (33,326)
Bad Recommendation	22% (9,775)
Unsure	3% (1,333)

2.4 Measuring Effectiveness of Feedback Mechanisms

We developed a custom machine learning model to estimate the similarity between the “rejected” video (i.e., the video a user clicked the button on) and subsequent recommended videos. The model required high-quality, labeled examples of “similar” and “not similar” video pairs for training and validation. Because there is no authoritative or automated ground truth for whether a recommendation is “bad” in the context of our study, we employed a team of 25 research assistants (RAs) to label video pairs according to our policy document defining criteria for “similarity” (e.g., related subject matter or viewpoints; see Appendix). Between April 22 and June 26, 2022, they classified 44,434 pairs of videos into Acceptable Recommendations, Bad Recommendations, and Unsure. Each video pair was labeled by a single RA to balance cost and throughput. This approach introduces potential annotator-specific variation; however, this classification is inherently subjective, and perfect inter-rater agreement is not possible. To mitigate individual biases, we randomized assignment of pairs to annotators, ensuring that idiosyncratic labeling tendencies averaged out across the dataset.

For the initial days of classification, we asked the research assistants to classify random pairs from our data. For the majority of the classification period, however, we asked them to classify pairs selected to have a wide range of different predicted (by our model) probabilities of being bad. This method, known as active learning, involved asking assistants to classify pairs where our model was uncertain, that is, its prediction probabilities were close to 0.50. Multiple rounds of the active learning cycle were conducted, during which the model was re-trained with freshly human-classified pairs that had previously been uncertain. Then a new set of uncertain predictions was sent to the research assistants for classification, leading to the next cycle of model retraining and calibration. In a deviation from typical active learning approaches, we had some cycles in which probabilities other than 0.5 were targeted in order to ensure calibration across a wide range of predicted probabilities.

From the 44,434 video pairs assessed by the research assistants, 75% were classified as Acceptable Recommendations, 22% as Bad Recommendations, and 3% as Unsure (see Table 1).

Operational definitions. If the RA judged the recommended video to be “too similar” to the rejected video, based on content, theme, or topic, it was marked as a **Bad Recommendation**. Otherwise, it was marked as an **Acceptable Recommendation**. Examples:

- **Bad Recommendation example:** A user rejects a video titled “*Why Climate Change is a Hoax*” from Channel A, and is later recommended “*The Truth About Global Warming Myths*” from Channel B. Both videos deny mainstream climate science and present similar argumentative content.

- **Acceptable Recommendation example:** A user rejects a video titled “*Why Climate Change is a Hoax*”, and is later recommended “*How Solar Panels Work*”. While both relate to the environment, the second video is educational and does not share the rejected video’s conspiratorial framing.

Bad Recommendation Rate. The proportion of video pairs in each experimental arm classified as Bad Recommendations determined the **Bad Recommendation Rate** for that arm.

Rejected video: A video on which the participant clicked the “Stop Recommending” button.

Video pair: A rejected video and a subsequent recommended video.

Bad Recommendation: A video pair deemed too similar according to human-labeled training data and model predictions.

Bad Recommendation Rate: The proportion of video pairs in each experimental arm classified as Bad Recommendations.

2.5 Data

We analyzed data collected from December 2, 2021 to June 26, 2022. In this time period there were 22,838 participants that opted into our experiment and data collection. After removing exceptional behavior (e.g. bots, scripts), we have 22,722 participants for analysis. Of these, 6,147 of them rejected at least one video. There were a total of 30,314 rejected videos. This underscores the rarity of explicit user feedback on platforms like YouTube, and the importance of analyzing it at scale.

Our participants were recommended 567,880,195 videos, and in total we were able to analyze 162,983,496 video pairs. Video metadata (title, description, transcript, and channel information) was acquired through automated web requests to YouTube. Due to the cost of proxy infrastructure required for these requests, we sampled approximately 29% of total recommendations at random for metadata collection. The same sampling proportion applies to rejected videos, yielding roughly 8,700 rejected videos with full metadata available for similarity analysis. Additionally, some videos were unavailable due to being age restricted, private, unlisted, or removed after being tagged.

2.6 Analysis

The labeled dataset was used to train a machine learning model (see Section 3) to estimate similarity for all 567 million video pairs, enabling us to compute Bad Recommendation rates at scale.

Bad Recommendation rates

As we mentioned before, a “Bad Recommendation rate” is defined as how often YouTube recommends videos that are similar to a video that a participant has rejected. The analysis of these rates is a measure of the effectiveness of YouTube’s user control mechanisms in preventing unwanted recommendations. The underlying belief is that, since YouTube recommends using user control mechanisms to manage recommendations, that using such mechanisms to express what types of recommendations are unwanted should be effective in preventing such recommendations.

As such, if negative feedback is submitted for a particular “rejected” video, we consider that any subsequent recommendation similar to the rejected video is “bad”. Of course, there is a lot of nuance here: it is not clear that a single negative feedback submission

should suppress all future recommendations about a topic, and it is also possible that the user behavior on YouTube after feedback submission may justify recommendations on topics similar to the video for which feedback was submitted. Regardless, as these are the only tools YouTube offers to prevent unwanted recommendations, we do expect them to be effective for that purpose.

We calculate the Bad Recommendation rate by pooling all video pairs within each experimental arm and determining the proportion classified as bad by our model, rather than averaging rates per user. This approach weights users proportionally to their activity. We tested alternative aggregation strategies (e.g., computing per-user or per-rejected-video rates and then averaging), but found no meaningful differences in the findings. Confidence intervals reported in figures are 95% bootstrap confidence intervals.

Interaction Rates

We calculate the interaction rate for participants as the sum of user control interactions made divided by number of videos watched. A user control interaction may be a press of the “Stop recommending” button, or a native YouTube control (Dislike, Not interested, Remove from watch history, or Don’t recommend channel). For the interaction rate analysis, we divide participants into two groups, those in the UX Control group (who have no “Stop recommending” button) and others (who do have it). We calculate the average interaction rate for each of these two groups. In the first group, the rate includes only native interactions, as they are the only options available, while the latter group includes “Stop recommending” button interactions.

3 Semantic Similarity Model

We developed a machine learning model to quantitatively measure similarity between pairs of YouTube videos, facilitating large-scale evaluation of YouTube’s responsiveness to user feedback.

3.1 Feature Extraction

Due to computational constraints, we relied on textual metadata rather than raw video or audio data, extracting: Video Title, Channel ID, Description, and Transcript (when available). Separate models were trained for cases with and without transcripts.

3.2 Model Architecture

We developed a unified architecture that uses cross-encoder language models. Cross-encoders take two texts together (e.g. two titles, two transcripts) and directly estimate their similarity. In contrast to bi-encoders, which independently transform each piece of text into an embedding, cross-encoders often achieve higher accuracy, especially when fine-tuned for a specific domain. We create separate cross-encoders for titles, descriptions, and transcripts. Each cross-encoder produces a scalar similarity score between 0 and 1. These scores are then combined with a channel-matching feature (1 if the two videos come from the same channel, 0 otherwise) and fed into a simple linear output layer (equivalent to logistic regression). Training this entire pipeline end-to-end on our labeled video pairs allows the cross-encoders for each text type to be fine-tuned in a domain-specific way, even though the original language models were trained on more general data. A diagram of the unified model is shown in Figure 1.

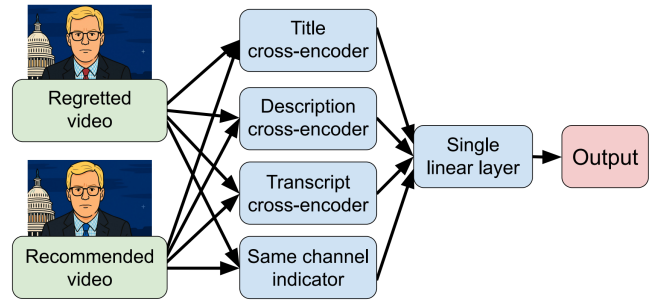


Figure 1: Diagram of the video similarity neural network.

Table 2: Performance metrics for both models.

Metric	Model with transcripts	Model without transcripts
Accuracy	91%	91%
AUROC	97%	97%
Precision	92%	95%
Recall	91%	87%

Because our dataset spans more than 100 languages, we use the mmarco-mMiniLMv2-L12-H384-v1 [46, 47], which supports a set of 100 languages. While preliminary checks suggested that performance is broadly consistent across most languages in our dataset, variations in label quality and the complexity of different languages could introduce uneven performance.

We do expect that bad YouTube recommendation rates may vary between different languages and geographic regions. We have chosen not to report any analysis of differences in bad recommendation rates between languages and countries, as we are not confident that any such findings are due to real differences in YouTube’s recommender system rather than a product of our model’s more subtle inconsistent behavior between different languages.

3.3 Training Procedure and Evaluation

Our labeled dataset (44,434 pairs, after removing the 3% of pairs labeled “unsure”) was split into training and testing sets (90%/10%). We used standard training protocols for language model fine-tuning, applying early stopping when improvements on a validation set plateaued. We did not see validation loss increase nor validation AUROC decrease during training, suggesting that the model was not overfitting. The final model demonstrated high performance, achieving AUROC scores of 97%, accuracy of 91%, precision between 92% and 95%, and recall of 87–91% (Table 2). It is difficult to compare these metrics to other methods as most other work is designed for video retrieval rather than directly assessing similarity, but these metrics by far surpassed our expectations.

4 Results

4.1 Interaction Rates

Our one-click button reduced friction relative to YouTube’s multi-step controls and increased feedback submission. Participants that see our button submit approximately 80 pieces of feedback per 1000 videos watched, versus an average of 37/1000 for users in our UX Control group, that did not see our button. That said, this difference should be interpreted cautiously, keeping in mind the large variation between participants.

4.2 Control Effectiveness

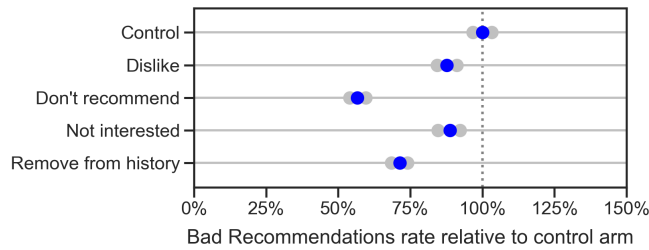


Figure 2: Overall Bad Recommendation rates per experiment arm, with 95% bootstrap confidence intervals. Rates are computed by pooling all video pairs within each arm. The percentages at the bottom represent rates relative to the Control arm, which is also indicated by the dashed vertical line.

For this analysis, we looked at the different feedback signals users send YouTube. In Figure 2 we see Bad Recommendation rates for the five different experiment groups: control, “Dislike”, “Don’t recommend”, “Not interested”, and “Remove from history”.

For people in the control group, the overall Bad Recommendation rate was about 2.3% – meaning that roughly one in every 43 recommendations was a Bad Recommendation. After calculating the Bad Recommendations rate for each experiment arm, we find that using the “Dislike” and “Not interested” buttons seemed to slightly decrease the Bad Recommendation rate, showing respective mean bad recommendation rates of 87.6% and 88.8% relative to the control group – a mere reduction of 10-11% on the baseline rate. Therefore, we can say these control mechanisms are marginally effective, having a very small impact on bad recommendations. The “Don’t recommend channel” and “Remove from history” buttons had slightly greater effectiveness, showing lower recommendation rates of 56.6% and 71.3% relative to the control group (corresponding reductions of 43.4% and 28.7% on the baseline rate).

4.3 Temporal Dynamics of Recommendations

To assess whether the effectiveness of user feedback changes over time, we examined the delay between a rejection event and subsequent recommendations. For each rejected video, we grouped recommendations by the number of days since the rejection and calculated Bad Recommendation rates within these time windows.

We observed that the highest Bad Recommendations rates occur immediately after feedback is given. This is shown in Figure 3. In the first week, Bad Recommendation rates remain close to the baseline

for each experimental arm, suggesting that YouTube’s recommendation system does not instantly incorporate negative feedback. Over the following days, the rates generally decline, indicating some gradual adjustment in the recommendation algorithm.

Still, even after four weeks, none of the feedback mechanisms fully eliminated Bad Recommendations, and the relative ordering of effectiveness between arms (*Don’t Recommend Channel* > *Remove from History* > others) remains consistent over time. In some cases – particularly for *Dislike* and *Not Interested* – Bad Recommendation rates returned to near-baseline levels after a week, suggesting that any algorithmic suppression of similar content was short-lived.

These results indicate that while certain controls can reduce exposure to similar content over longer time horizons, their immediate impact is limited, and effects can decay relatively quickly. The persistence of unwanted recommendations over time further underscores the need for more reliable user control mechanisms.

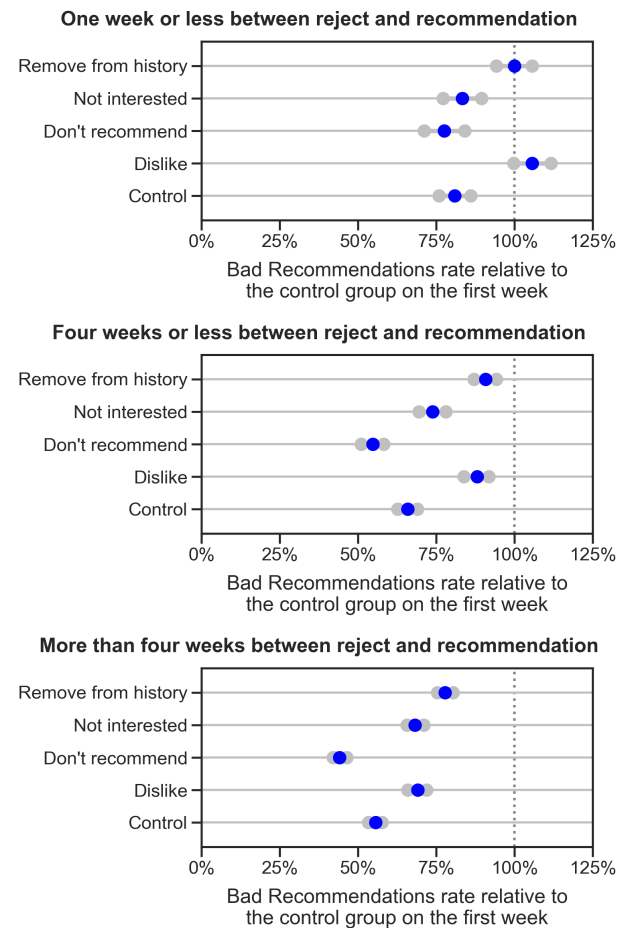


Figure 3: Bad Recommendation rates over time after a rejection event, grouped by experiment arm, with 95% bootstrap confidence intervals. The plot shows that the immediate effect of feedback is limited, with gradual decline in Bad Recommendations over the following days, and varying degrees of persistence across feedback types.

4.4 Impact of Channel

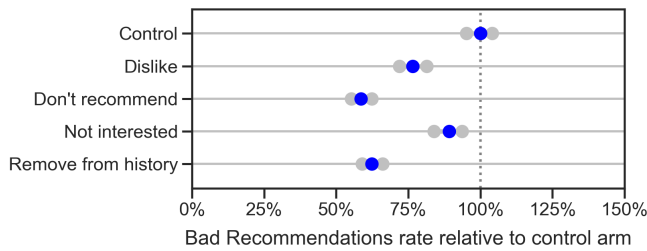


Figure 4: Bad Recommendation rates among pairs in which rejected and recommended videos come from different channels, by experiment arm, with 95% bootstrap CI.

For this analysis, we looked at only those video pairs where the rejected and recommended videos came from different channels, as shown in Figure 4. We find that, in this case, the “Don’t recommend channel” button is still the most effective, with an average bad recommendation rate of 58.6% relative to the control group (i.e. a reduction of 41.4% on the original rate). This may reflect fewer exposures to that channel and/or YouTube interpreting the signal as general negative feedback.

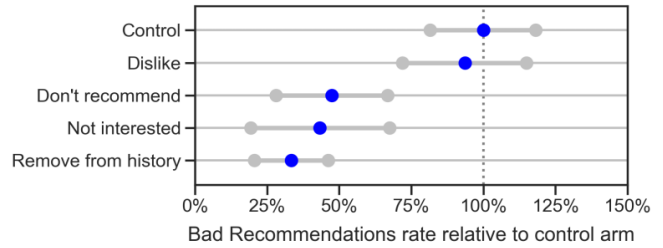
Compared to the other user controls we tested, people might have a clear idea about what the “Don’t recommend channel” button is meant to do – block a channel from recommendations. We can analyze whether the same channel continues to pop up after a participant clicks this button. For people in our control group, about 0.4% of subsequent recommendations after a rejected video were from the same channel as the rejected video. Meanwhile in the “Don’t recommend channel” group, we see this rate drop to about 0.1%. In other words, telling YouTube to stop recommending a channel seems to have the impact you might expect in most cases: fewer videos from that channel.

However, this behavior does not appear to be consistent. In about 0.1% of the video pairs we analyzed, a recommendation was made from the same channel. Even when we limited our analysis to just a one-week time period between rejected video and recommendation, we saw the same pattern persist. Our data does not allow us to confirm that the participant has not given YouTube a reason to ignore the “Don’t recommend channel” feedback, perhaps watching a video from that channel that shows up in a search result, but it seems unlikely that this explains all the cases that we see.

4.5 Analysis: User Interface

We also find that user feedback seems slightly more effective for homepage than sidebar recommendations, for almost all arms of the experiment, as shown on Figure 5. One possible interpretation could be that sidebar recommendations are made in a higher-information context (YouTube knows what video you are currently watching) and so it is easier to optimize for engagement and there is less weight given to user feedback. For homepage recommendations, YouTube has less ability to know what might be engaging at the moment and so gives more weight to user feedback.

Homepage:



Sidebar:

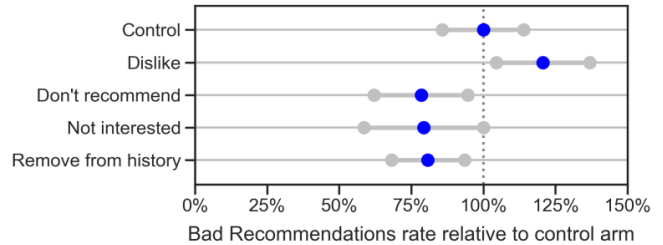


Figure 5: Bad Recommendation rates per experiment arm and source of recommendation, with 95% bootstrap CI.

5 Discussion and Conclusion

Our controlled experiment shows that while YouTube’s user controls (e.g., “Dislike,”) exhibit some impact, none reliably prevent unwanted recommendations. Despite the platform’s claims, our data suggests that user feedback plays a limited role in shaping the recommendation algorithm – an issue compounded by opaque design choices and engagement-driven optimization.

Our findings relate to and extend prior work such as Liu, Wu, and Resnick’s controlled experiment on training YouTube recommenders to avoid unwanted videos [27], which finds that repeatedly using *Not Interested* can reduce recommendations on a targeted topic by up to 88%. We observe substantially weaker reductions for *Not Interested* (and *Dislike*). We view these results as complementary rather than contradictory. While their study examines a smaller number of sock-puppet accounts under tightly controlled conditions, our approach operates at a much larger scale, with over 22,000 participants generating more than 567 million recommendation observations in the wild. In our setting, feedback is sparse (about 30,000 rejections across 22,000 users), users interact across diverse topics and sessions, and subsequent behavior may reintroduce similar content into recommendations. Moreover, our outcome measures the persistence of *content-similar* recommendations following a specific rejection, rather than suppression of a single predefined topic. Together, these differences help explain the gap in effect sizes and suggest that prior audits provide an upper bound on feedback effectiveness, whereas our results reveal how these mechanisms perform in practice.

Our findings complement earlier qualitative reports from the Mozilla Foundation [28], which documented instances of recommendations which were harmful and/or misaligned with YouTube’s own guidelines. In our view, this misalignment between user intent and recommendations reflects a deeper disconnect between user expectations and algorithmic incentives. Tools like “Dislike” and

“Not Interested” often fail to deliver meaningful change, and user interfaces remain cumbersome. A baseline rate of one unwanted recommendation per roughly 43 total recommendations might seem low in absolute terms, particularly as a YouTube video will get less than 30 recommendations, and a user might only look at the top ones. However, a rate of $1/43 \approx 2.3\%$ is a meaningful rate given the scale of exposure of users to the YouTube recommender system.

5.1 Model Bias Considerations

Our similarity model, trained on pre-existing language data, may encode societal biases that influence similarity scores. While our randomized experimental design mitigates systematic bias across treatment groups, we did not systematically assess biases for every language in our dataset, and hence we refrain from drawing conclusions about differences in YouTube’s recommender performance across languages. Future work should investigate these fairness concerns, possibly incorporating additional data (e.g. video topic).

5.2 Limitations

We use content similarity as a proxy for undesired recommendations. While scalable and systematic, it is not a perfect reflection of each user’s intent in context. People may reject videos for reasons unrelated to their content (e.g., timing, mood, or video length), and they may welcome videos that are similar in topic but differ in tone or framing. Conversely, our similarity model may classify two videos as dissimilar despite them being undesirable to the user for other reasons. These mismatches mean that our Bad Recommendation Rate should be interpreted as an approximate indicator rather than an exact measure of user satisfaction.

A second limitation is that we cannot account for user-driven behaviors that might reintroduce similar content into their recommendations. For example, a participant might reject a video about a topic but later search for the same topic, which could signal renewed interest to the recommendation algorithm. Our data collection focuses on recommendations surfaced in standard browsing contexts (homepage, sidebar, watched videos) and does not include search queries or direct navigation. As such, some of the “persistence” we observe in unwanted recommendations may reflect user-driven re-engagement rather than the platform disregarding feedback.

5.3 Policy Recommendations

5.3.1 Make user controls understandable and accessible. Research into recommender systems has shown that people feel more satisfied when controls are paired with the ability to make meaningful choices about the recommendation algorithm [14]. Users need clear, in-product explanations of how controls influence recommendations, not just policy overviews. For example, “Don’t Recommend Channel” sends a stronger signal than “Dislike,” but this distinction is not communicated. Under the EU’s Digital Services Act (DSA) [5], such transparency will soon be mandatory, and YouTube should ensure these obligations are met through improved interface design.

5.3.2 Empower users to shape recommendations. Current feedback tools are reactive and insufficient. Our data shows that people continue receiving unwanted content, including borderline or potentially harmful videos, even after submitting negative feedback.

Recommendation systems should prioritize user-defined preferences and provide options for users to proactively specify interests and exclusions. Simply put, “asking people what they want instead of just watching what they do” [43].

5.3.3 Improve data access for researchers. Researchers need to be able to access a platform in order to adequately study it. YouTube’s public and researcher-facing APIs remain limited in scope, and crowdsourced tools like RegretsReporter exist because platforms restrict researcher access. The European Digital Media Observatory (EDMO) has proposed a code of conduct on how platforms can give researchers data while protecting user privacy, in compliance with the General Data Protection Regulation (GDPR) [32]. Mandating researcher access to platform data was one key outcome of the EU Digital Services Act (DSA) adopted by the European Parliament in July 2022. Under the DSA’s article on “Data Access and Scrutiny”³, vetted independent researchers must be given the tools they need to investigate and run experiments on recommender systems as they relate to platform harms or systemic risks.

5.3.4 Protect public-interest research. We relied on a crowdsourced approach to studying YouTube for this study, working with tens of thousands of volunteers who offered up their data to support the research. Without access to data from real people about what they are experiencing, it’s impossible to analyze the spread and impact of algorithmic harms on the platform. Policymakers should introduce protections for ethical platform audits, including data donation, scraping, and “sock puppet” methods, provided they adhere to privacy standards. This would ensure civil society actors can investigate platforms without fear of reprisal.

5.4 Conclusion

The content that users consume on a platform is the output of a complex feedback loop between user preferences and choices, platform algorithms, and the content available at the time, and it is not trivial to separate each factor. The few studies that do attempt to do so often focus on a single step [10, 19, 26, 29, 31, 35, 38].

YouTube should treat feedback as a meaningful signal, not a secondary input to engagement metrics. Designing better feedback mechanisms could be a win-win for both users and the platform: rather than only making choices on the user’s behalf, and producing mechanisms of addiction and polarization, recommender systems can promote pluralism and user empowerment [18]. People could have a greater say over what they see, and YouTube wouldn’t have to rely as heavily on passive data collection in order to infer what people want to watch.

Our findings show that YouTube’s user controls are insufficient, inconsistently effective, largely reactive, and poorly communicated. YouTube should adopt stronger transparency mechanisms, redesign its feedback tools, and provide data access to enable accountability. Only then can platforms begin to align their recommendation engines with the well-being and preferences of their users.

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³See Article 40 of the DSA report [5] (previously Article 31).

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A Appendix: RegretsReporter Classification Policy

A.1 Background

The RegretsReporter extension adds a “STOP RECOMMENDING” button to YouTube videos. Here, pressing it on a video will be referred to as “regretting” that video. This is intended to express to YouTube that the user does not wish to receive recommendations similar to the regretted video.

A.2 Task

The task is to classify pairs of videos, one of which was regretted, and the other is a recommendation received after the regret. We want to assess whether that recommendation was in violation of a reasonable interpretation of the participants’ wishes expressed by the regret. We refer to the pair as *violative* or *non-violative*.

A.3 Goals

For the classification task in general, the goals are:

- Find examples of violative pairs or sets of violative pairs from a participant that tell a compelling story of YouTube not adequately respecting user preferences. This lack of respect could be intentional (e.g., prioritizing engagement over respect for feedback) or unintentional (e.g., they are unable to recognize that two videos are similar).
- Allow us to compute a proportion of recommendations that are violative for each participant, allowing us, through our experimental design, to evaluate the effectiveness of each of YouTube’s user control mechanisms.

A.4 Classification Policy

The research assistant should indicate that a recommended video is “acceptable” due to it being adequately unrelated to the regretted video, or indicate that it is “unacceptable” or that the two videos are too similar.

In the latter case, the research assistant will also choose one from a list of possible reasons for which the videos are similar, or select “other” and provide a written description. Finally, the research assistant will also choose whether to flag a recommendation as potentially distressing or harmful.

A.5 Use of Data

The data you provide will be used to:

- Fulfill the two goals listed above.
- Provide data to build and evaluate a machine learning algorithm which will aim to identify YouTube regrets in a larger dataset.

- Identify features of pairs that are difficult for machine learning models to detect. It is possible that there are types of pairs for which it is difficult to detect that they are violative using a machine learning model. As YouTube certainly applies such methods, it’s possible that YouTube will make violative recommendations at a high rate in these sorts of cases, and that our model may be unable to detect that these pairs are violative.

A.6 Possible Manners of Similarity

A.6.1 Similar Subject Matter. Two videos are similar due to subject matter if they are about the same thing, to the finest level of granularity that a general audience would be aware of. For example:

- Two videos about the same sport team are similar, but two videos about different teams in the same sport are not.
- Two videos about the same video game are similar, even if they are regarding different aspects of the game.
- Two videos about the same film are similar.
- Two music videos are similar if the songs are by the same artist(s).

A.6.2 Similar Opinion on Divisive Social or Political Issue. If a participant regrets a video with a clear opinion on a divisive social or political issue, any other video with the same opinion is unacceptable. For example:

- A video that criticizes one group of people (nationality, political orientation, etc.) is similar to any other video that criticizes the same group.
- A video that argues for a political position is similar to any other video that argues for the same position.

A.6.3 Relating to the Same Controversy of Limited Interest. Two videos relating to a controversy not of general interest are similar. This is distinct from the above, as even if videos have different opinions, the recommendation may be unacceptable if it not an issue of general interest (e.g. national and global issues such as immigration, unemployment, the environment) and is plausible that the user simply does not want to hear about the issue at all. Examples of such issues might be celebrity scandals or debates about the result of a football game.

A.6.4 Featuring the Same Persons. If the primary host or featured person of two videos is the same, they are considered to be similar.

A.6.5 Belong to the Same Class of Potentially Undesirable Content. If the regretted video belongs to a class of potentially undesirable content, any other video with the same class of content is considered similar. For example:

- Videos featuring graphic injury or violence
- Videos featuring sexual content
- Videos featuring unpleasant language
- Videos featuring hate speech

A.6.6 Other. If a pair of videos does not satisfy any of these categories, but the research assistant still thinks they should be classified as similar, the option “Other” will allow for a written description of why that is the case. Otherwise, that pair of videos should be classified as not similar to each other.

B RegretsReporter Onboarding Flow

B.1 Welcome!

RegretsReporter will help you take control of your YouTube recommendations.

It is also part of Mozilla’s crowdsourced research into YouTube’s algorithm to learn more about what people regret being recommended and how the algorithm responds to user feedback.

B.2 What happens if you participate?

- If you choose to participate in our research, you may be assigned to a **“control” (or “placebo”) group** and the **extension may not function**.
- See more information about our research here.
- If you do choose to participate, please leave the extension installed even if it does not appear to be doing anything.

B.3 Data Collection

If you participate in our research, the extension will send Mozilla information about:

- The number of videos you view
- Your use of features to control your recommendations
- Information about the videos that you are recommended

To learn more about what RegretsReporter will collect, read our Privacy Notice.

B.4 Consent Checkbox

- ☐ **I would like to participate in Mozilla’s research into YouTube’s algorithmic controls.**

By checking the box above and submitting, you consent to participate in the study and agree to the described data collection and experiment assignment process.

C Crowdsourced research info

When you install this extension you are asked to participate in a randomized experiment to support Mozilla’s research into YouTube’s algorithmic controls. This document describes the experiment in more detail.

C.1 Frequently Asked Questions

C.1.1 Why don’t I see the “STOP RECOMMENDING” button? The button is not available for playlist recommendations, it is only available for individual video recommendations.

If you opted in to participating in Mozilla’s research you may have been assigned to a control group for which the button is not available. In this case, in order to support our research, we ask that you leave the extension installed for at least one month and continue to use YouTube as normal.

C.1.2 I have pressed the “STOP RECOMMENDING” button but it does not seem to be changing my recommendations - why is this? If you opted in to participating in Mozilla’s research, you may have been assigned to a control group for which the button is available but has no effect. In this case, in order to support our research, we ask that you leave the extension installed for at least one month and continue to use YouTube as normal.

It is also possible that YouTube is not acting on the feedback signals sent by pressing our button. By continuing to contribute data to our research, you can help us measure this.

C.1.3 What happens if I do not opt in to the experiment? If you do not opt in to the experiment, the randomized assignment will not occur and no data will be sent to Mozilla. Pressing the “STOP RECOMMENDING” button will send a YouTube “dislike” message. For the purposes of extension code, you will be considered to be in the dislike-and-collection-opt-out experiment arm.

C.2 Experiment Arms

If you agree to participate, you will be randomly assigned to one of these experiment arms. Please do not uninstall the extension if you end up in an arm you don’t like, as this will impact our research. Read our Privacy Notice to see what data Regrets Reporter collects.

- **no-action:** Pressing the “STOP RECOMMENDING” button will have no effect.
- **dislike:** Pressing the “STOP RECOMMENDING” button will send a YouTube “dislike” message.
- **remove-from-history:** Pressing the “STOP RECOMMENDING” button will send a message to remove the video from your watch history if possible, otherwise will send a YouTube “dislike” message.
- **not-interested:** Pressing the “STOP RECOMMENDING” button will send a YouTube “Not interested” message if possible, otherwise will send a YouTube “dislike” message.
- **dont-recommend:** Pressing the “STOP RECOMMENDING” button will send a YouTube “Don’t recommend channel” message if possible, otherwise will send a YouTube “dislike” message.
- **no-ui-data-collection:** The “STOP RECOMMENDING” button will not be shown, but standard data will still be collected.
- **dislike-and-collection-opt-out:** Pressing the “STOP RECOMMENDING” button will send a YouTube “dislike” message. No data will be collected or sent to Mozilla.